

ASSIGNMENT 3

QUESTION 1

hostname	ipconfig	getmac	ping	arp
nbstate	route	path	pathping	netstat
tracert				

Practice the commands mentioned in the above table and fill in the blanks.

SOLUTION

Command	Working
ping	Command is one of the most often used networking utilities for detecting devices on a network and for troubleshooting network problems.
hostname	command that displays the hostname of your machine
ipconfig	frequently used utility that is used for finding network information about your local machine like ip addresses, dns addresses etc.
nbstat	diagnostic tool for troubleshooting netbios problems
netstat	used for displaying information about tcp and udp connections and ports.
getmac	command that shows the mac address of your network interfaces
arp	for showing the address resolution cache. this command must be used with a command line switch -a is the most common.
tracert	command prints the path. if all routers on the path are functional, this command prints the full path.
path	command specifies the location where ms-dos should look when it executes a command.
route	command allows you to make manual entries into the network routing tables.
pathping	after sending out packets from you to a given destination, it analyzes the route taken and computes packet loss on a per-hop basis.

Outputs

ipconfig

txt

```
1 Windows IP Configuration
2
3
4 Ethernet adapter Ethernet 2:
5
6     Connection-specific DNS Suffix  . :
7     Link-local IPv6 Address . . . . . : fe80::5ca6:b97c:1b7f:ce94%9
8     IPv4 Address. . . . . : 172.16.119.58
9     Subnet Mask . . . . . : 255.255.255.0
```

```
10 Default Gateway . . . . . : 172.16.119.1
11
12 Wireless LAN adapter Local Area Connection* 1:
13
14 Media State . . . . . : Media disconnected
15 Connection-specific DNS Suffix . :
16
17 Wireless LAN adapter Local Area Connection* 6:
18
19 Media State . . . . . : Media disconnected
20 Connection-specific DNS Suffix . :
21
22 Wireless LAN adapter Wi-Fi 2:
23
24 Media State . . . . . : Media disconnected
25 Connection-specific DNS Suffix . :
```

ping

txt

```
1 Pinging google.com [172.217.26.14] with 32 bytes of data:
2 Reply from 172.217.26.14: bytes=32 time=10ms TTL=117
3 Reply from 172.217.26.14: bytes=32 time=10ms TTL=117
4
5 Ping statistics for 172.217.26.14:
6     Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
```

hostname

txt

```
1 DESKTOP-IDSI54N
```

netstat

txt

```
1 Active Connections
2
3 Proto Local Address Foreign Address State
4 TCP 127.0.0.1:49674 DESKTOP-IDSI54N:49675 ESTABLISHED
5 TCP 127.0.0.1:49675 DESKTOP-IDSI54N:49674 ESTABLISHED
6 TCP 127.0.0.1:59005 DESKTOP-IDSI54N:59436 ESTABLISHED
7 TCP 127.0.0.1:59006 DESKTOP-IDSI54N:59008 ESTABLISHED
8 TCP 127.0.0.1:59008 DESKTOP-IDSI54N:59006 ESTABLISHED
9 TCP 127.0.0.1:59333 DESKTOP-IDSI54N:59005 TIME_WAIT
10 TCP 127.0.0.1:59372 DESKTOP-IDSI54N:59005 TIME_WAIT
11 TCP 127.0.0.1:59394 DESKTOP-IDSI54N:59005 TIME_WAIT
12 TCP 127.0.0.1:59419 DESKTOP-IDSI54N:59005 TIME_WAIT
13 TCP 127.0.0.1:59436 DESKTOP-IDSI54N:59005 ESTABLISHED
14 TCP 172.16.119.58:58959 a23-208-178-166:https ESTABLISHED
```

getmac

txt

```
1 Physical Address Transport Name
2 =====
3 D8-9C-67-6F-4E-1F Media disconnected
4 F4-39-09-22-9E-A7 \Device\Tcpip_{65BECBCF-235F-424E-82A7-FE2902C3E542}
5
6 arp -a
7 Interface: 172.16.119.58 --- 0x9
```

8	Internet Address	Physical Address	Type
9	172.16.119.1	74-8e-f8-90-80-78	dynamic
10	172.16.119.21	f4-39-09-22-9e-68	dynamic
11	172.16.119.27	f4-39-09-22-9d-b8	dynamic
12	172.16.119.54	f4-39-09-22-9f-2a	dynamic
13	172.16.119.88	f4-39-09-22-9d-c9	dynamic
14	172.16.119.89	f4-39-09-22-9f-15	dynamic
15	172.16.119.90	f4-39-09-22-9e-08	dynamic
16	172.16.119.91	f4-39-09-22-9d-ac	dynamic
17	172.16.119.95	f4-39-09-22-9d-8e	dynamic
18	172.16.119.96	f4-39-09-22-9e-73	dynamic
19	172.16.119.101	f4-39-09-22-9d-ab	dynamic
20	172.16.119.102	f4-39-09-22-9d-cf	dynamic
21	172.16.119.105	f4-39-09-22-9d-d9	dynamic
22	172.16.119.255	ff-ff-ff-ff-ff-ff	static
23	224.0.0.2	01-00-5e-00-00-02	static
24	224.0.0.22	01-00-5e-00-00-16	static
25	224.0.0.251	01-00-5e-00-00-fb	static
26	224.0.0.252	01-00-5e-00-00-fc	static
27	239.255.250.250	01-00-5e-7f-fa-fa	static
28	239.255.255.250	01-00-5e-7f-ff-fa	static

arp -a

txt

1	Interface: 172.16.119.58 --- 0x9		
2	Internet Address	Physical Address	Type
3	172.16.119.1	74-8e-f8-90-80-78	dynamic
4	172.16.119.21	f4-39-09-22-9e-68	dynamic
5	172.16.119.27	f4-39-09-22-9d-b8	dynamic
6	172.16.119.54	f4-39-09-22-9f-2a	dynamic
7	172.16.119.88	f4-39-09-22-9d-c9	dynamic
8	172.16.119.89	f4-39-09-22-9f-15	dynamic
9	172.16.119.90	f4-39-09-22-9e-08	dynamic
10	172.16.119.91	f4-39-09-22-9d-ac	dynamic
11	172.16.119.95	f4-39-09-22-9d-8e	dynamic
12	172.16.119.96	f4-39-09-22-9e-73	dynamic
13	172.16.119.101	f4-39-09-22-9d-ab	dynamic
14	172.16.119.102	f4-39-09-22-9d-cf	dynamic
15	172.16.119.105	f4-39-09-22-9d-d9	dynamic
16	172.16.119.255	ff-ff-ff-ff-ff-ff	static
17	224.0.0.2	01-00-5e-00-00-02	static
18	224.0.0.22	01-00-5e-00-00-16	static
19	224.0.0.251	01-00-5e-00-00-fb	static
20	224.0.0.252	01-00-5e-00-00-fc	static
21	239.255.250.250	01-00-5e-7f-fa-fa	static
22	239.255.255.250	01-00-5e-7f-ff-fa	static

tracert -h 5 google.com

txt

1	Tracing route to google.com [142.251.221.238]				
2	over a maximum of 5 hops:				
3					
4	1	<1 ms	2 ms	1 ms	172.16.119.1
5	2	<1 ms	<1 ms	<1 ms	172.31.1.6
6	3	20 ms	25 ms	36 ms	112.196.126.1

```
7 4 5 ms 4 ms 4 ms 202.164.32.25
8 5 4 ms 4 ms 4 ms 202.164.51.36
9
10 Trace complete.
```

path

txt

```
1 PATH=C:\ProgramData\Oracle\Java\javapath;C:\Program Files (x86)\Intel\Intel(R)
Management Engine Components\iCLS\;C:\Program Files\Intel\Intel(R) Management
Engine Components\iCLS\;C:\WINDOWS\system32;C:\WINDOWS;C:
\WINDOWS\System32\Wbem;C:\WINDOWS\System32\WindowsPowerShell\v1.0\;C:
\WINDOWS\System32\OpenSSH\;C:\Program Files (x86)\Intel\Intel(R) Management
Engine Components\DAL;C:\Program Files\Intel\Intel(R) Management Engine
Components\DAL;C:\Program Files (x86)\Intel\Intel(R) Management Engine
Components\IPT;C:\Program Files\Intel\Intel(R) Management Engine
Components\IPT;C:\Program Files\dotnet\;C:
\Users\CSED\AppData\Local\Microsoft\WindowsApps;
```

routeprint

txt

```
1 =====
2 Interface List
3  9...f4 39 09 22 9e a7 .....Intel(R) Ethernet Connection (7) I219-LM #2
4  17...da 9c 67 6f 4e 1f .....Microsoft Wi-Fi Direct Virtual Adapter #5
5  10...fa 9c 67 6f 4e 1f .....Microsoft Wi-Fi Direct Virtual Adapter #6
6  1.....Software Loopback Interface 1
7  14...d8 9c 67 6f 4e 1f .....Realtek RTL8821CE 802.11ac PCIe Adapter #2
8  =====
9
10 IPv4 Route Table
11 =====
12 Active Routes:
13 Network Destination    Netmask          Gateway          Interface        Metric
14 0.0.0.0                0.0.0.0          172.16.119.1     172.16.119.58    281
15 127.0.0.0              255.0.0.0        On-link          127.0.0.1        331
16 127.0.0.1              255.255.255.255  On-link          127.0.0.1        331
17 127.255.255.255        255.255.255.255  On-link          127.0.0.1        331
18 172.16.119.0           255.255.255.0    On-link          172.16.119.58    281
19 172.16.119.58          255.255.255.255  On-link          172.16.119.58    281
20 172.16.119.255         255.255.255.255  On-link          172.16.119.58    281
21 224.0.0.0              240.0.0.0        On-link          127.0.0.1        331
22 224.0.0.0              240.0.0.0        On-link          172.16.119.58    281
23 255.255.255.255        255.255.255.255  On-link          127.0.0.1        331
24 255.255.255.255        255.255.255.255  On-link          172.16.119.58    281
25 =====
26 Persistent Routes:
27 Network Address        Netmask  Gateway Address  Metric
28 0.0.0.0                0.0.0.0  172.16.119.1     Default
29 =====
30
31 IPv6 Route Table
32 =====
33 Active Routes:
34 If Metric Network Destination    Gateway
35 1 331 ::1/128                On-link
36 9 281 fe80::/64              On-link
```

```
37  9    281 fe80::5ca6:b97c:1b7f:ce94/128
38                                     On-link
39  1    331 ff00::/8                    On-link
40  9    281 ff00::/8                    On-link
41  =====
42 Persistent Routes:
43  None
```